

1	Sized to 1~2m span, to eventually carry a 100gram jumping mechanism.
2	Propulsion efficiency optimal at Strouhal number ~0.3
3	Chord ~0.2m, wing motion frequency ~5Hz
4	Need low KV Motor with around 20~60W cruise power.
5	Lightweight motor has KV>3000, this requires a gear reduction ratio 30~40:1
6	Considering worm drive
7	Dismissing worm drive for low efficiency (~0.30). Use planar gear
8	Industrial 48DP Acetal gear have teeth 18~120
9	No more than 2 reduction stages. All horizontal rod should not rotate and should be used as structure rod.
10	Use industrial 48DP 18T pinion and 120T gear
11	Industrial gears have ID5/32 bore
12	Common Carbon Fiber rod has OD1/8
13	OD1/8 rod has a terrible ±0.006" tolerance. After sanding the rod, the rod still cannot fit the bearing
14	Tapering the OD1/8 carbon fiber with a lathe. Failed. Dismiss OD1/8 carbon fiber rod
15	Consider using OD1/8 aluminum rod
16	OD1/8 aluminum rod not strong enough according to crash estimation. Consider using good surface finishing OD1/4 carbon fiber tube
17	OD1/4 tube requires ID1/4 bearing, which is available
18	ID1/4 bearing has at least 3/8 gear bore
19	32DP gear has OD3/8 gear bore. Check compatibility of 32DP gear
20	Dismiss 32DP gear for low compactness. Stick to 48DP gear. Eventually discover the hobbyist world, where 48DP ID3/8 gear is a standard
21	Check whether I can combine a 18Tpinion and a 98T gear into a compound gear
22	Failed. Failed. Failed. Failed.
23	Managed to combine a 30T pinion and a 98T gear into a well-concentric compound gear
24	Reduction ratio around 24:1. Not low enough.
25	Extensively searching for lightweight motor with KV<2200
26	Find a rare lightweight KV<2200 motor. Try to taper its output shaft to fit the pinion
27	Failed
28	Find a light weighted KV~1600 motor with ~50W cruise power. Perfect. Also figure out a way to fit its 3mm metric shaft into the 1/8in pinion. See the website on tolerance.
29	Refine kinematics to reach ~0.30 Strouhal number
30	Assemble the gearbox

